ECD Mapping: Pulse Realms — Team Arena

Game Overview

- **Mechanic:** 3v3 team combat where every attack/heal/shield is gated behind an MCQ
- **Engine:** KAPLAY.js
- **Subject:** Math & Science (configurable via `SUBJECT_IDS`)
- **Duration:** 5 min per match (`matchDurationSec: 300`)
- **Roles:** Attacker, Healer, Builder (each with unique abilities)

---

1. Competency Model

What skills and knowledge does this game assess?

Content Knowledge (D1)

Competency	Description	Curriculum Alignment
Mathematical Fluency	Speed and accuracy solving math problems under time pressure	CCSS.Math operations, number sense
Scientific Reasoning	Applying science concepts to answer questions correctly	NGSS disciplinary core ideas
Adaptive Knowledge	Maintaining accuracy as difficulty increases (adaptive engine raises difficulty after 3 correct in a row)	Cross-curricular metacognition

21st-Century Skills (D3–D5)

Competency	Description
Strategic Thinking	Selecting appropriate roles and actions based on game state
Teamwork & Prosocial Behavior	Choosing to heal/shield allies vs. focusing on damage
Emotional Regulation	Maintaining performance under combat pressure (low HP, time pressure)
Persistence	Continuing to engage after incorrect answers
Risk Assessment	Choosing between safe actions (shield) and high-risk/high-reward actions (power strike)

---

2. Evidence Model

What observable in-game behaviors indicate competency?

Content Knowledge Evidence

Observable Behavior	Telemetry Event	Payload Fields	Evidence Rule
Answers question correctly	`question_answered`	`correct: true`	Direct evidence of content mastery for the question's `subject` and `difficulty`
Answers quickly and correctly	`question_answered`	`correct: true, responseTimeMs < 3000`	Evidence of fluent, confident knowledge (not guessing)
Answers quickly but incorrectly	`question_answered`	`correct: false, responseTimeMs < 2000`	Evidence of guessing behavior — no mastery
Answers slowly but correctly	`question_answered`	`correct: true, responseTimeMs > 5000`	Evidence of effortful recall — knowledge present but not fluent
Maintains accuracy at higher difficulty	`question_answered`	`difficulty: 4–5, correct: true`	Evidence of deep understanding beyond surface recall
Accuracy improves within session	sequence of `question_answered`	`correct` trend over `ts`	Evidence of learning during gameplay

Strategic Behavior Evidence

Observable Behavior	Telemetry Event	Payload Fields	Evidence Rule
Chooses healer role	session `meta`	`role: 'healer'`	Preference for support/prosocial play style
Uses diverse action types	`action_performed`	varying `actionType`	Higher Shannon entropy = strategic flexibility
Targets low-HP enemies	`action_performed`	`targetId` cross-ref with game state	Evidence of tactical awareness
Switches from attack to heal when ally is low	`action_performed` sequence	`actionType` change pattern	Evidence of situational awareness and teamwork

SEL / Affective Evidence

Observable Behavior	Telemetry Event	Payload Fields	Evidence Rule
Continues playing after 2+ wrong answers	`question_answered` sequence	`correct: false` followed by further actions	Persistence after failure — growth mindset indicator

Accuracy drops when HP is low	`question_answered` + `damage_taken`	`correct` rate when cumulative `damage_taken.amount` > 72 (60% of 120 HP)	Low emotional regulation under stress
Response time increases after failure	`question_answered` sequence	`responseTimeMs` after `correct: false`	Possible frustration or increased deliberation (context-dependent)
Heals teammates before self-interest actions	`action_performed`	`actionType: 'heal'` targeting allies when attacker role could deal damage	Empathy / prosocial prioritization

---

3. Task Model

What game tasks elicit the evidence above?

Core Task Loop

```
1. Player selects target → chooses action (attack/heal/shield/build)
2. Question engine selects MCQ at adaptive difficulty
   - Difficulty = base (3 + action's difficultyBias) adjusted by player streak
   - After 3 correct → difficulty +1; after 2 wrong → difficulty -1
   - Questions drawn from subject-specific JSON pools, avoiding recently seen
3. Player answers within 8-second time limit (questionTimeLimitSec)
4. If correct: action fires with speed-based power multiplier
   - multiplier = max(1, min(2, 2.2 - responseTimeMs/5000))
   - Fast + correct = up to 2× power; slow + correct = 1× power
5. If incorrect: action misses (effectType: 'miss')
6. Combat effects applied (damage/heal/shield)
7. Match runs for 300 seconds or until objective captured
```

Task Design Constraints

Design Feature	Purpose	Config Reference
8-second time limit	Prevents unlimited deliberation; creates authentic pressure	`constants.js:4 — questionTimeLimitSec: 8`
Adaptive difficulty (1-5 scale)	Keeps questions in Zone of Proximal Development	`questionEngine.js:51-56 — computeDesiredDifficulty()`
Speed-based power multiplier	Incentivizes confident, fluent knowledge (fast + correct = more damage)	`combatSystem.js:4-6 — speedMultiplier()`
Role selection (3 roles)	Forces strategic choice; reveals preferred play style	`constants.js:33-37 — ROLE_IDS`
6 action types per role	Ensures diverse evidence collection across action categories	`constants.js:39-46 — ACTION_IDS`

No respawn (`respawnEnabled: false`)	Raises stakes — mistakes have consequences, elicits emotional regulation	`constants.js:22`
3v3 team format	Creates social context for prosocial behavior measurement	AI teammates in Phase 1

---

4. Assembly Model

How does evidence combine into competency scores?

Per-Session Aggregation

```
For each game session:
  1. Collect all question_answered events
  2. Compute:
     - correctness_rate = correct / total
     - avg_response_time = mean(responseTimeMs)
     - speed_accuracy_profile = classify(dominant response pattern)
     - action_variation_index = shannon_entropy(actionType distribution)
     - persistence = actions_after_failure / total_failures
     - frustration_score = accuracy_when_low_hp vs accuracy_when_healthy
     - learning_velocity = accuracy_second_half - accuracy_first_half
  3. Store in computed_metrics table
```

Cross-Session Aggregation (Daily Rollup)

```
For each student per day:
  1. Average computed_metrics across all sessions
  2. Track trends:
     - accuracy_trend = linear regression slope over last 7 days
     - engagement_trend = session_count change week-over-week
     - mastery_by_subject = per-subject correctness_rate
  3. Store in student_metrics_daily table
```

Mastery Thresholds

Level	Correctness Rate	Response Profile	Interpretation
Not Yet	< 0.40	Any	Significant knowledge gaps; needs targeted support
Developing	0.40–0.65	Mostly deliberate or struggling	Knowledge present but not fluent; benefits from practice
Proficient	0.65–0.85	Mix of fluent and deliberate	Solid understanding; can handle grade-level content

Advanced	> 0.85	Mostly fluent	Confident mastery; ready for acceleration

Adaptive Difficulty Response

The question engine (in `questionEngine.js:51–56`) uses a built-in assembly model:

- `skillRating` starts at 3 (mid-range)
- After each answer: `skillRating += correct ? 0.15 : –0.2` plus accuracy adjustment
- `correctStreak >= 3` → requested difficulty +1
- `wrongStreak >= 2` → requested difficulty -1
- This creates a real-time competency estimate that adapts task difficulty accordingly

---

References

File	Relevant Content
`PulseRealms/src/systems/combatSystem.js`	Telemetry events emitted (lines 71-113)
`PulseRealms/src/systems/questionEngine.js`	Adaptive difficulty algorithm (lines 51-56, 92-111)
`PulseRealms/src/systems/telemetry.js`	Event storage format (lines 35-42)
`PulseRealms/src/config/constants.js`	Game configuration values
`EdGame Analytics Blueprint.md`	6-dimension taxonomy and ECD framework